

Saehoon Eo

Dept. of Mathematics, Stanford
eosehoon@stanford.edu
Last update: July 22, 2026

EDUCATION

Department of Mathematics, Stanford University

Sep. 2024 – Present

Stanford, CA, United States

- Third-year Ph.D. Student
- Advisor: Prof. Jonathan Luk

Department of Mathematical Sciences, KAIST

Mar. 2020 – Aug. 2024

Daejeon, Republic of Korea

GPA: 4.28/4.3

- B.S. in Mathematical Sciences (Summa Cum Laude, Valedictorian)
- Advisor: Prof. Moon-Jin Kang
- Left for mandatory military service (Jul. 2021 – Jan. 2023)

Korea Science Academy of KAIST

Mar. 2017 – Feb. 2020

Busan, Republic of Korea

GPA: 4.28/4.3

- Government-run gifted high school
- Graduated with honors: Valedictorian, Top Honors in Mathematics

RESEARCH INTEREST

- Partial Differential Equations, Kinetic Theory, General Relativity

AWARDS & HONORS

Dean's List, Department of Mathematical Science, KAIST

Fall 2023, Spring 2023, Spring 2021

Honor Student, Qualification for taking high-level graduate courses

Spring 2021 – Spring 2024

Required to take 85 credits within 6 semesters with GPA higher than 3.7

Korea Presidential Science Scholarship, KSAF

Spring 2020 – Spring 2024

Scholarship for honorable undergrad. in science from the Korean government

KAIST Presidential Fellowship (KPF), KAIST

Spring 2020 – Spring 2024

Granted to 30 undergraduates in about 700 freshmen, join KPF honor society

Graduation Award: Top Prize in Mathematics, KSA of KAIST

Feb. 2020

Distinction Award, Singapore International Mathematics Challenge

May 2018

RESEARCH PUBLICATION

- **Eo, S.** (2026). Stability of the Schwarzschild Spacetime in the Einstein--Vlasov System via Nonlinear Phase Mixing. *In preparation*.
- **Eo, S., Eun, N., Kang, M.** (2026). Global Existence of Classical Solutions to Brenner-Navier-Stokes-Fourier System for Large Data. *Preprint, arXiv:2605.07350*.
- **Eo, S., Eun, N.** (2025). Traveling wave solutions to a large class of Brenner-Navier-Stokes-Fourier system. *Comm. Math. Sci.*, 24(6), 1577-1593 (2026)
- **Eo, S., Eun, N., Kang, M., Oh, H.** (2025). Stability of Riemann shocks for isothermal Euler by inviscid limits of global-in-time large Navier-Stokes flows. *Preprint, arXiv:2505.15078*.

- **Eo, S., Eun, N., Kang, M. (2024).** Stability of a Riemann shock in a physical class: from Brenner-Navier-Stokes-Fourier to Euler. *Preprint, arXiv:2411.03613*.
- **Eo, S., Eun, N., Kang, M., Oh, H. (2025).** Traveling wave solutions to Brenner-Navier-Stokes-Fourier system. *J. Differ. Equat.*, 422, 639–658.
- **Eo, S., Hwang, S., Yeo, W. (2020).** Group Rings Satisfying Nil Clean Property. *Communications of the Korean Mathematical Society*, 35(1), 117–124.

ACADEMIC ACTIVITY

PDE Perspectives in Gravitational Theory International Centre for Mathematical Sciences	Sep. 07 – Sep. 11, 2026 <i>Edinburgh, UK</i>
Thematic Program on Shocks and Singularities The Fields Institute	May. 11 – May. 22, 2026 <i>Toronto, ON, Canada</i>
3rd Simons Math Summer Workshop: Partial Differential Equations of Classical Physics Simons Center for Geometry and Physics	Jul. 7 – Jul. 25, 2025 <i>Stony Brook, NY, United States</i>
Workshop on Kinetic Theory and Fluids University of Wisconsin-Madison	Mar. 28 – Mar. 30, 2025 <i>Madison, WI, United States</i>
New Trends in Hyperbolic Conservation Laws and Related Models Courtyard Marriott Namdaemun	Jun. 23 – Jun. 29, 2024 <i>Seoul, Korea</i>
The 10th Korea PDE Winter School Korea Advanced Institute of Science and Technology	Feb. 19 – Feb. 23, 2024 <i>Daejeon, Korea</i>
Fluids in Seoul 2024 June E Huh Center for Mathematical Challenges/KIAS	Jan. 15 – Jan. 19, 2024 <i>Seoul, Korea</i>
IAS/PCMI Undergraduate Summer School Park City Mathematics Institute — Institute for Advanced Study	Jul. 16 – Aug. 5, 2023 <i>Park City, UT, United States</i>
Singapore International Mathematics Challenge NUS High School of Math & Science	May 21 – May 25, 2018 <i>Singapore, Singapore</i>

TEACHING EXPERIENCE

Teaching Assistant for Math 21: Calculus	Winter 2026
Course Assistant for Math 205A: Real Analysis	Fall 2025
Course Assistant for Math 175: Elementary Functional Analysis	Spring 2025
Course Assistant for Math 115: Functions of a Real Variable	Fall 2024
Tutor for Freshmen: MAS101 Calculus I	Spring 2023

TALKS

Stability of the Schwarzschild Spacetime in the Einstein-Vlasov System via Nonlinear Phase Mixing KAIST, Department of Mathematical Sciences, Partial Differential Equation Seminar	Jul. 22, 2026
Stability of the Schwarzschild Spacetime in the Einstein-Vlasov System via Nonlinear Phase Mixing KIAS, School of Mathematics, Analysis, PDE & Probability Seminar	Jul. 13, 2026

PRESENTATIONS & GROUP SEMINARS

Speaker of Student Geometry Seminar A Scattering Theory Approach to Mass Inflation	May. 04, 2026
Speaker of Student Analysis Seminar Noether's Theorem and Conservation Laws on Linear Schrödinger Equations	Apr. 24, 2026
Speaker of Student Geometry Seminar Stability of Catenoid for HVMC Equation (Part 6)	Apr. 06, 2026
Speaker of Student Geometry Seminar Stability of Catenoid for HVMC Equation (Part 3)	Feb. 02, 2026
Speaker of Student Geometry Seminar Burnett's Conjecture in Generalized Wave Coordinates (Part 3)	Dec. 01, 2025
Speaker of Student Analysis Seminar Vector Field Method and ILED on Nonlinear Schrödinger Equations	Nov. 21, 2025
Speaker of Student Analysis Seminar Noether's Theorem and Conservation Laws on Linear Schrödinger Equations	Oct. 24, 2025
Speaker of Student Analysis Seminar Local Existence of Compressible Euler Equations	Apr. 25, 2025
Speaker of Student Analysis Seminar Dynamical Optimal Transport	Feb. 7, 2025
Speaker of Math Club Help Desk Session for Students Preparing for Exams, MAS101 Calculus I, KAIST	Apr. 7, 2024
Presentation at KAIST Math Problem Solving Club Compressible Euler Equations	Nov. 27, 2023
Speaker of Department Help Desk MAS441: Lebesgue Integral Theory, KAIST	Oct. 12, 2023
Presentation at IAS/PCMI Summer School Presentation on the Research Project at IAS/PCMI Summer School	Aug. 4, 2023
Organizer of a Reading Seminar on Functional Analysis Covered <i>Functional Analysis, Sobolev Spaces and Partial Differential Equations</i> by H. Brezis	Spring 2023

OTHER EXPERIENCE

Department's Student Council — Executive Member - Organize Department Career Seminar for Undergraduates, Fall 2023 - Organize Department Open Lab Event, Fall 2023/Spring 2024 - Mentor for Sophomores, Spring 2023	Spring 2023 – Spring 2024
Republic of Korea Army, 9th div.	Jul. 5, 2021 – Jan. 4, 2023
KAIST Undergraduate Math Colloquium — Member of Operating Committee	Spring 2021
Summer Camp 2019 <i>Jacobs University Bremen — VDI</i> - A week of science camp with international participants - Financially supported by KSA and Ministry of Science and ICT in South Korea	Jul. 2019 <i>Bremen, Germany</i>